12: Sampling Distribution and Central Limit Theorem(CLT)

Principles of Datascience with R

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PSTAT 10

Research question

- Every data science project aims to answer a research question (or multiple questions).

- **Example**
  - Do UCSB students who _____ have higher GPA?

Important Terminology

- A population is the full group we want to understand

- A sample is a subset of that population that we have/collect data for.
  - The penguins dataset was a sample from the population

Important Terminology

- A population is the full group we want to understand

- Population Parameter is a number associated with the population
  - Population mean μ
  - Population standard deviation σ

Important Terminology

- A sample is a subset of that population that we have data for

- Sample Statistic is number calculated from the sample
  - Sample Mean $\bar{x}$
  - sample median,
  - sample variance s^2,
  - Sample standard deviation s

A statistic can be used as an estimate of a parameter

Descriptive Statistics vs. Inferential Statistics

- In descriptive statistics (in EDA), we use **sample statistics** such as the sample mean or proportion to understand/summarize the observed data.

- In inferential statistics we use the observed data(sample) to make an inference(a conclusion) about the **population parameters** using probabilistic models (Module 2).

Parameter estimation

- We are often interested in *population parameters*.
- Since complete populations are difficult (or impossible) to collect data on, we use *sample statistics* as *point estimates* for the unknown population parameters of interest.
- But sample statistics vary from sample to sample.
- Quantifying how much sample statistics vary provides a way to estimate the *margin of error (also known as standard error)* associated with our point estimate.
- But before we get to quantifying the variability among samples, let's try to understand how and why point estimates vary from sample to sample.

Suppose we randomly sample 1,000 adults from each state in the US a few times.

Would you expect the sample means of their heights to be the same, somewhat different, or very different?

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Would you expect the sample means of their heights to be the same, somewhat different, or very different?

Not the same, but only somewhat different.

Suppose the proportion of all American adults who support the expansion of solar energy is $p = 0.88$, which is our parameter of interest. Is a randomly selected American adult more or less likely to support the expansion of solar energy?

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Yes, more likely.

Suppose that you don't have access to opinion (for or against solor power) of the entire population of all American adults, which is a quite likely scenario.

In order to estimate the proportion of American adults who support solar power expansion, you might sample from the population and use your sample proportion as the best guess for the unknown population proportion.

- Sample, with replacement, 1000 American adults from the population, and record whether they support solar power or not expansion.
- Find the sample proportion.
- Do this many times
- Plot the distribution of all of the sample proportions

```
# 1. Create a set of 250 million entries, where 88% of
# them are "support" and 12% are "not".

pop_size <- 250000000
possible_entries <- c(rep("support", 0.88 * pop_size),
                      rep("not", 0.12 * pop_size))

# 2. Sample 1000 entries without replacement.

sampled_entries <- sample(possible_entries, size = 1000)

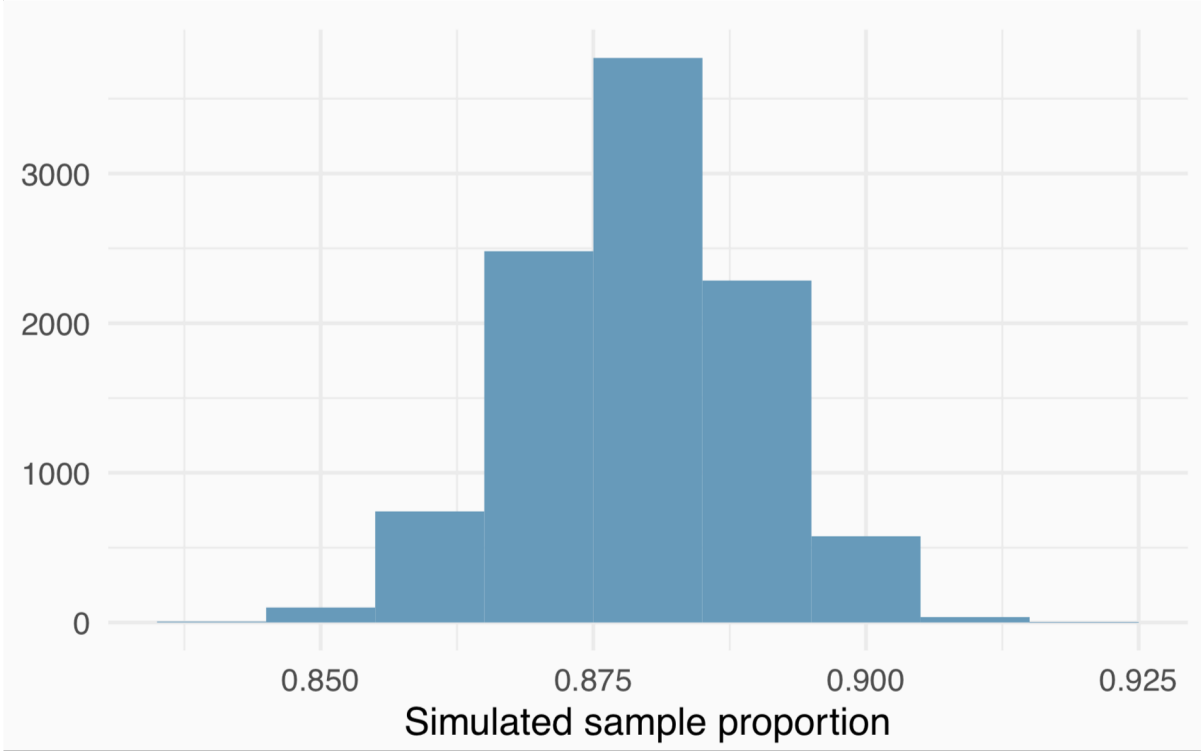
# 3. Compute p-hat: count the number that are "support",
# then divide by # the sample size.

sum(sampled sampled_entries == "support") / 1000

# Note: This gives one sample proportion value.
```

Sampling distribution (of sample proportion)

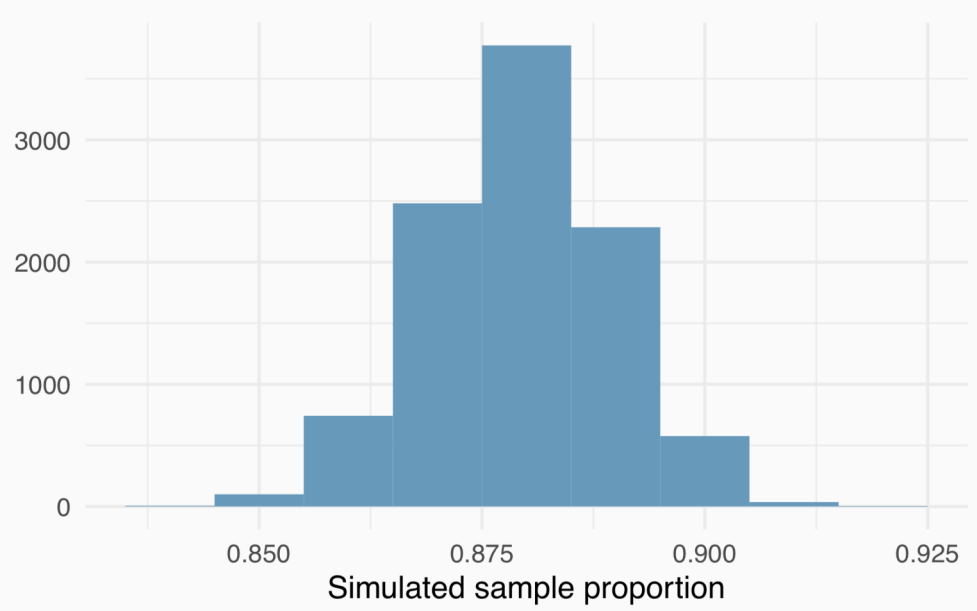
Suppose you were to repeat this process many times and plot the results. What you just constructed is called a simulated sampling distribution.



Sampling distribution (of sample proportion)

What is the shape and center of this distribution?

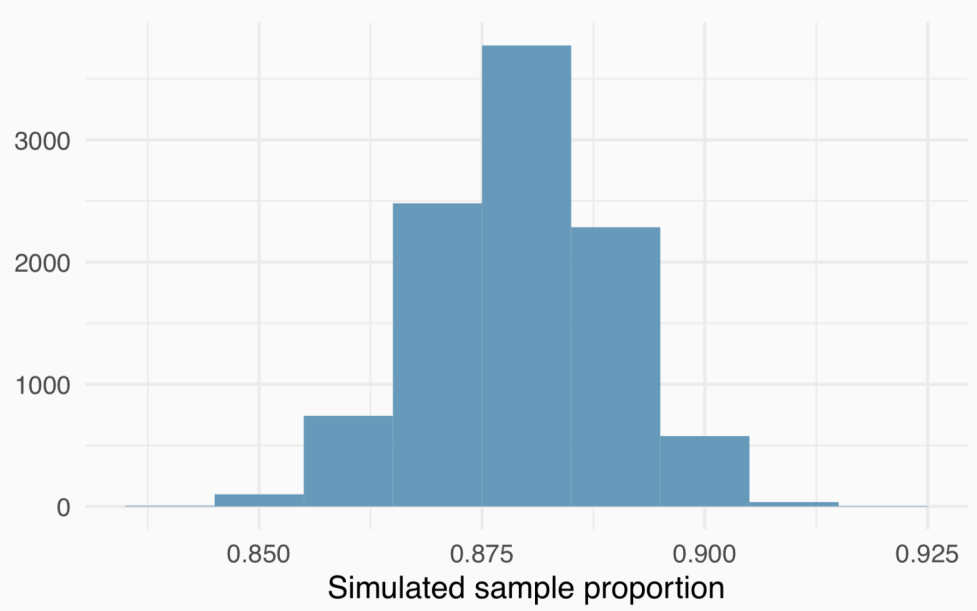
The distribution looks symmetric and somewhat bell-shaped.



Sampling distribution (of sample proportion)

Based on this distribution, what do you think is the true population proportion?

The center of the distribution: about 0.88.



Note: 0.88 was the true population proportion for the population from which we drew our samples for the sampling distribution!

And we recovered it from the center of the sampling distribution!

Sampling distribution (of a Statistic)

is the Probability Distribution of a Statistic under repeated sampling

- Values of a statistic vary because random samples vary (ie A statistic is a random variable)
- "Sampling distribution" or "probability distribution" of the statistic:
  - All possible values of the statistic,
  - and all the corresponding probabilities
- Can be hard to calculate
  - Either have to do the math
  - Or have to generate all possible samples and calculate the statistic based on each sample

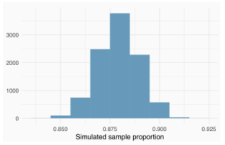
Simulated or Empirical Distribution (of a Statistic)

- Simulated distribution of the statistic:
  - Based on simulated values of the statistic
  - Consists of all the observed values of the statistic,
  - and the proportion of times each value appeared

- Good approximation to sampling distribution
  - if the number of repetitions in the simulation is large

Sampling distributions are never observed

- In real-world applications, we only have one sample
- So, we never actually observe the sampling distribution, yet it is useful to always think of a point estimate as coming from such a hypothetical sampling distribution.



- Understanding the sampling distribution will help us characterize and make sense of the point estimates that we do observe.

Sampling Distribution and Central Limit Theorem

Sampling Distribution and Central Limit Theorem

- The Central Limit Theorem describes how the normal distribution is connected to sampling distributions of sample averages.
- We care about sample averages because they estimate population averages
- [Remember: proportions are a special case of averages.]

Central Limit Theorem

If the sample size is

- large, and
- sample is drawn at random with replacement(independent draws)

Then, *regardless of the distribution of the population*,

the probability distribution of the sample average (sampling distribution)

is roughly normal

Note: CLT also holds for proportions

Exactly Which Normal distribution?

The sampling distribution (distribution of the sample averages and proportions) is roughly normal

- centered at the population average μ.
- SD = (population SD) / √sample size

$$= \frac{\sigma}{\sqrt{n}}$$

Th SD of the sampling distribution is called the standard error

Why Is This Important?

- The center of the sampling distribution captures the unknown population average μ
  - Sample average can be used as an estimate of population average
  - The variability of the sampling distribution helps us measure how accurate the sample average is as an estimate of the population average.

Young, Underemployed and Optimistic

Coming of Age, Slowly, in a Tough Economy

Young adults hit hard by the recession. A plurality of the public (41%) believes young adults, rather than middle-aged or older adults, are having the toughest time in today's economy. An analysis of government economic data suggests that this perception is correct. The recent indicators on the nation's labor market show a decline in the

Tough economic times altering young adults' daily lives, long-term plans. While negative trends in the labor market have been felt most acutely by the youngest workers, many adults in their late 20s and early 30s have also felt the impact of the weak economy. Among all 18- to 34-year-olds, fully half (49%) say they have taken a job they didn't want just to pay the bills, with 24% saying they have taken an unpaid job to gain work experience. And more than one-third (35%) say that, as a result of the poor economy, they have gone back to school. Their personal lives have also been affected: 31% have postponed either getting married or having a baby (22% say they have postponed having a baby and 20% have put off getting married). One-in-four (24%) say they have moved back in with their parents after living on their own.

http://pewresearch.org/pubs/2191/young-adults-workers-labor-market-pay-careers-advancement-recession

Standard Error (SE) (Margin of error)

The general public survey is based on telephone interviews conducted Dec. 6-19, 2011, with a nationally representative sample of 2,048 adults ages 18 and older living in the continental United States, including an oversample of 346 adults ages 18 to 34. A total of 769 interviews were completed with respondents contacted by landline telephone and 1,279 with those contacted on their cellular phone. Data are weighted to produce a final sample that is representative of the general population of adults in the continental United States. Survey interviews were conducted under the direction of Princeton Survey Research Associates International, in English and Spanish. Margin of sampling error is plus or minus 2.9 percentage points for results based on the total sample and 4.4 percentage points for adults ages 18-34 at the 95% confidence level.

41% ± 2.9%: We are 95% confident that 38.1% to 43.9% of the public believe young adults, rather than middle-aged or older adults, are having the toughest time in today's economy.

49% ± 4.4%: We are 95% confident that 44.6% to 53.4% of 18-34 years olds have taken a job they didn't want just to pay the bills.

41% ± 2.9% and 49% ± 4.4% are called 95% confidence intervals and are interval estimates for the unknown population proportions.

Discussion Question

After rolling 1,000,000 fair 6-sided dice, which of these histograms would you expect to have a bell shape? Check all that apply.

1) The histogram of outcomes of these million rolls
2) The histogram that results from computing the average outcome of these million rolls
3) The histogram that results from splitting the outcomes into 1,000 groups of 1,000 (in the order they occurred) and computing the average outcome of each group

Review for further understanding

Understand: Why is There a Distribution?

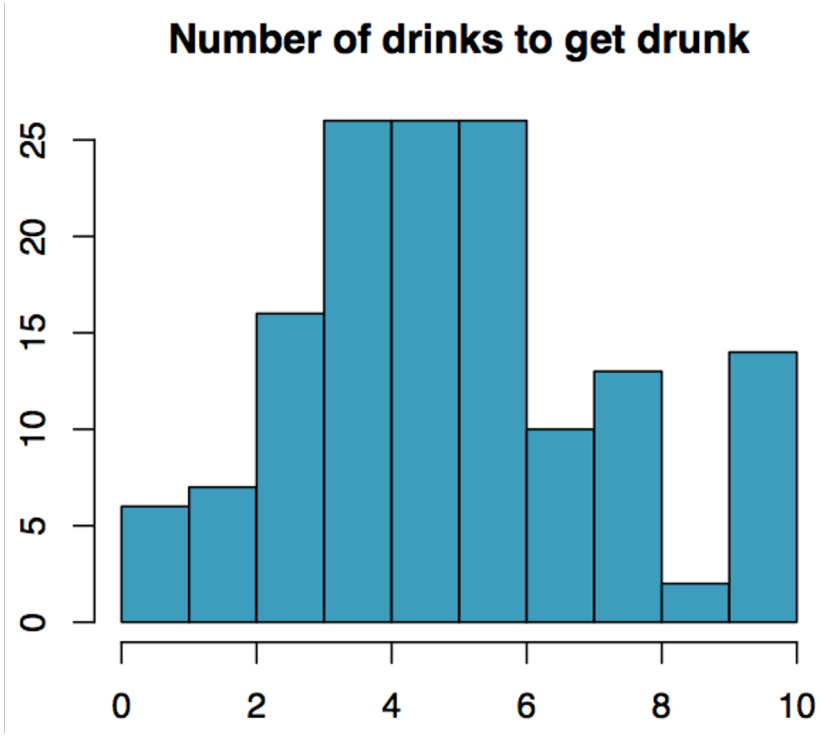
- You have only one random sample, and it has only one average.
- But **the sample could have come out differently**.
- And then the sample average might have been different.
- So there are many possible sample averages.

Distribution of the Sample Average

- Imagine all possible random samples of the same size as yours. There are lots of them.
- Each of these samples has an average.
- The **distribution of the sample average** is the distribution of the averages of all the possible samples.

Activity

The following histogram shows the distribution of number of drinks it takes a group of college students to get drunk. We will assume that this is our population of interest. If we randomly select observations from this data set, which values are most likely to be selected, which are least likely?



Suppose that you don't have access to the population data. In order to estimate the average number of drinks it takes these college students to get drunk, you might sample from the population and use your sample mean as the best guess for the unknown population mean.

- Sample, with replacement, ten students from the population, and record the number of drinks it takes them to get drunk.
- Find the sample mean.
- Repeat the above many times
- Plot the distribution of the sample averages obtained across many repetitions

1	7	16	3	31	5	46	4	61	10	76	6	91	4	106	6	121	6	136	6
2	5	17	10	32	9	47	3	62	7	77	6	92	0.5	107	2	122	5	137	7
3	4	18	8	33	7	48	3	63	4	78	5	93	3	108	5	123	3	138	3
4	4	19	5	34	5	49	6	64	5	79	4	94	3	109	1	124	2	139	10
5	6	20	10	35	5	50	8	65	6	80	5	95	5	110	5	125	2	140	4
6	2	21	6	36	7	51	8	66	6	81	6	96	6	111	5	126	5	141	4
7	3	22	2	37	4	52	8	67	6	82	5	97	4	112	4	127	10	142	6
8	5	23	6	38	0	53	2	68	7	83	6	98	4	113	4	128	4	143	6
9	5	24	7	39	4	54	4	69	7	84	8	99	2	114	9	129	1	144	4
10	6	25	3	40	3	55	8	70	5	85	4	100	5	115	4	130	4	145	5
11	1	26	6	41	6	56	3	71	10	86	10	101	4	116	3	131	10	146	5
12	10	27	5	42	10	57	5	72	3	87	5	102	7	117	3	132	8		
13	4	28	8	43	3	58	5	73	5.5	88	10	103	6	118	4	133	10		
14	4	29	0	44	6	59	8	74	7	89	8	104	8	119	4	134	6		
15	6	30	8	45	10	60	4	75	10	90	5	105	3	120	8	135	6		

Example

List of random numbers: 59, 121, 88, 46, 58, 72, 82, 81, 5, 10

1	7	16	3	31	5	46	4	61	10	76	6	91	4	106	6	121	6	136	6
2	5	17	10	32	9	47	3	62	7	77	6	92	0.5	107	2	122	5	137	7
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5	6	20	10	35	5	50	8	65	6	80	5	95	5	110	5	125	2	140	4
6	2	21	6	36	7	51	8	66	6	81	6	96	6	111	5	126	5	141	4
7	3	22	2	37	4	52	8	67	6	82	5	97	4	112	4	127	10	142	6
8	5	23	6	38	0	53	2	68	7	83	6	98	4	113	4	128	4	143	6
9	5	24	7	39	4	54	4	69	7	84	8	99	2	114	9	129	1	144	4
10	6	25	3	40	3	55	8	70	5	85	4	100	5	115	4	130	4	145	5
11	1	26	6	41	6	56	3	71	10	86	10	101	4	116	3	131	10	146	5
12	10	27	5	42	10	57	5	72	3	87	5	102	7	117	3	132	8		
13	4	28	8	43	3	58	5	73	5.5	88	10	103	6	118	4	133	10		
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15	6	30	8	45	10	60	4	75	10	90	5	105	3	120	8	135	6		

Sample mean: (8+6+10+4+5+3+5+6+6+6) / 10 = 5.9

Repeat many times and Plot all sample means

Sampling distribution

What you just constructed is called a *sampling distribution*.

Sampling distribution

What you just constructed is called a *sampling distribution*.

What is the shape and center of this distribution? Based on this distribution, what do you think is the true population average?

Acknowledgement

- Datascience in a box
- Data 8
- Open Intro Statistics